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TECHNOLOGY

What Blockchain Can't Do

by Catherine Tucker and Christian Catalini

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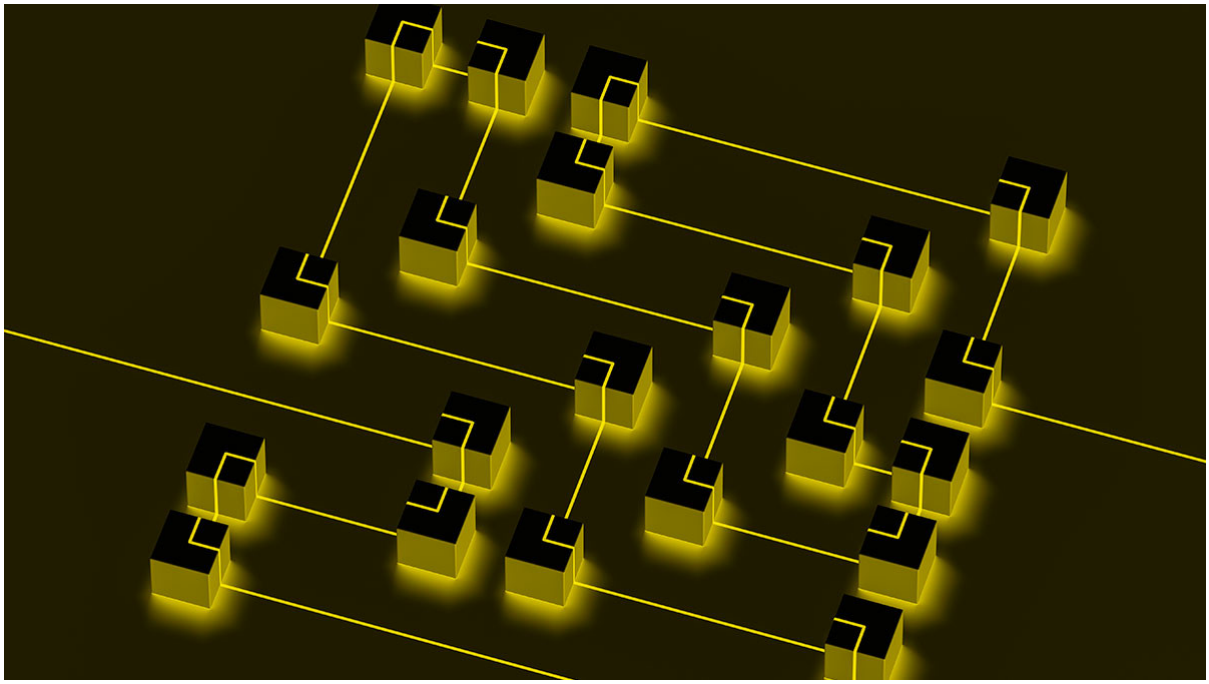

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Blockchain technology has the potential to do amazing things. It can provide an immutable, digital audit trail of transactions, and can be used to cheaply verify the integrity of data. It can help businesses and individuals agree, on a global scale, about the true state of affairs within a market without relying on a costly intermediary.

This is achieved through a clever combination of economic incentives and cryptography, and ensures that at any point in time, digital records reflect the true "consensus" among the key stakeholders involved. When it comes to sharing digital records and assets, it can therefore replace the need for trust between players, or the need for a central authority to verify and maintain the records of transactions.

However, when assessing blockchain business models, it is useful to understand what blockchain can't do.

Think about the problem of tracking babies within a hospital ward and beyond. This is a very serious problem. The consequences of a baby being mistaken for another baby can be horrendous. Therefore, storing records that contain a baby's current location in a way that makes these data points immutable and verifiable seems like a great use of blockchain technology.

But there is a big problem with using blockchain to solve such a problem. The digital records may be immutable and verifiable, but how does someone know which digital record is attached to which baby? To link an entry on the blockchain to an actual, real-life baby, we need to give the baby a physical identifier through a physical tag, or in a more futuristic world, a small chip or digital genome record that links the baby to its digital record. And this is where blockchain falls down. It can't help with this process, and can't verify that perhaps the most important step of verification is happening correctly.

At the interface between the offline world and its digital representation, the usefulness of the technology still critically depends on trusted intermediaries to effectively bridge the "last mile" between a digital record and a physical individual, business, device, or event. In our example, the technology would have to rely on humans to correctly and honestly implement the match between baby and digital record. And if humans get that wrong or manipulate the data when it is entered, in a system where records are believed ex-post as having integrity, this can have serious negative consequences.

On the other hand, if the link between an individual and their medical record is successfully established and the last-mile problem is solved, then a blockchain can be used not only to ensure data integrity but also to give individuals control over how their medical data is used (for academic research, a fitness app, or commercial drug development, for example).

There are other parallel examples. Within marketing, one issue that often comes up is that a pair of eyeballs that an advertiser is paying for may not actually belong to the person they're supposed to. The advertiser might think they're paying to show an ad to a mid-thirties male in the market for a Lamborghini, but the ad might actually be shown to a minivan-driving academic who has no intention of buying another car for kids to wreck but who likes to dream. Or, even worse, the ad could be being viewed by a bot. Blockchain technology can track which digital identifiers are associated with the viewing of an ad, but it cannot help with verifying humanness or the honesty of a buyer's intentions. Verifying who's actually behind the digital identifier requires offline verification. Verifying the honesty of apparent buying intentions is perhaps beyond any technology we possess today.

On the bright side, blockchain technology can be used to change the relationship between digital content creators, advertisers, and consumers. Advertisers can reward users for their attention by giving them access to exclusive online content they would otherwise have to pay for. Content creators can explore new monetization models that benefit from blockchain's ability to cheaply and effectively settle transactions. While consumers hate micropayments because of the mental costs they involve — micropayments are like a hated "taxi meter" in consumers' heads — this could reshape how paywalls and

subscriptions work behind the scenes across different digital properties. Moreover, if we are not worried about verifying a pair of eyeballs' humanness, but instead want to ensure ownership over digital records such as browsing data, then blockchain can work perfectly. One of the issues we face constantly in establishing the economics of privacy is the issue of property rights over data. And blockchain is perfectly positioned to define them.

As the ecosystem around blockchain technology develops, new types of intermediaries will emerge that turn the last-mile problem, of keeping digital records in sync with their offline counterparts, into actual business opportunities. While the technology is early stage, as these key complements mature, blockchain has the potential to fundamentally reshape ownership over digital data, and the digital platforms we use every day.

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sushil kumar 2 days ago

This article is all over the place. No clear outline of practical use cases, wherein Blockchain is applicable or not.

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